

The Observer and the World

Predictive Closure, the Projective Shadow, and a First-Principles Grammar of Empirical Science

PREDICTIVE CLOSURE · OBSERVATION GEOMETRY · RECONSTRUCTION BY FAILURE

Nature is not given to science from nowhere. It is encountered from somewhere.

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25 August 2026

Theorem 1 (Observation Projection). Let $z = (z_1, z_2)^\top \neq 0$ be a homogeneous two-channel state, let $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in GL(2, \mathbb{R})$, and suppose observation identifies every nonzero common rescaling $z \sim \lambda z$. On the chart $z_2 \neq 0$, with projective coordinate $q = z_1/z_2$, the induced observable transformation is

$$q' = \frac{aq + b}{cq + d}.$$

Hence the cross-ratio of every admissible quadruple is invariant. Conversely, if T is a continuous injective map of a real interval into the real projective line and preserves all admissible cross-ratios, then T is the restriction of a fractional-linear transformation and admits a two-dimensional homogeneous linear representative, unique up to multiplication by a nonzero scalar.

Proof. The forward direction follows by dividing the two components of Mz . For the converse, choose three distinct anchors. The unique Möbius transformation agreeing with T on those anchors has the same cross-ratio with every fourth point; because cross-ratio with three fixed distinct anchors is injective in the fourth coordinate, the two maps agree throughout the interval. The representing matrix is unique up to common nonzero scale. $\square$

THE PROJECTIVE SHADOW

Linear homogeneous source $\rightarrow$ scale-blind observation $\rightarrow$ fractional-linear shadow

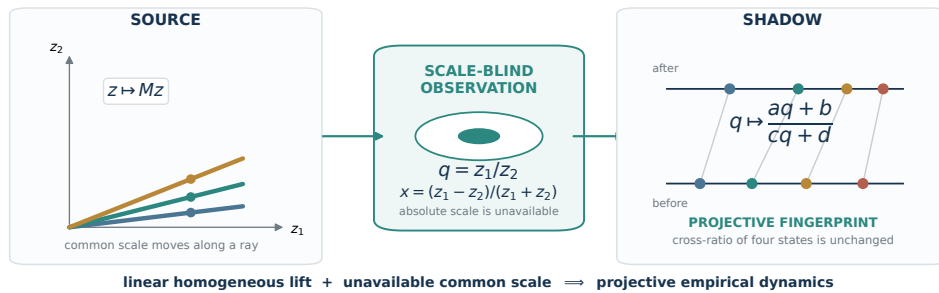


Figure 1: One exact branch of the observer-world interface. A homogeneous linear source evolves as $z \mapsto Mz$. If common scale is unavailable, the observer retains only a ray, ratio, or normalized contrast. The induced empirical law is fractional-linear and cross-ratio preservation is its observable fingerprint. The Projective Shadow is therefore not a universal geometry of observation; it is the geometry forced by this specific lens.

Abstract

Science is never practiced from nowhere. An observer is inside nature: it can intervene, record, and retain only distinctions available through a physical interface. This paper develops a first-principles grammar for what follows from that fact. Relative to a declared access layer, histories are identified when every admissible future test assigns them the same future law; the resulting predictive quotient is the coarsest state sufficient for that jurisdiction. Raw interventions can be reduced by the same criterion. State is therefore earned from surviving predictive distinctions before coordinates or geometry are assigned.

Within this architecture, Theorem 1 isolates an exact observation geometry. When common scale is unavailable and a homogeneous two-channel state transforms linearly, the observable ratio or normalized contrast necessarily transforms fractionally linearly and preserves cross-ratio. Conversely, continuous injective cross-ratio preservation on a real interval identifies a fractional-linear map and an abstract two-dimensional homogeneous lift. This is the **Observation Projection Theorem**; the induced law is the **Projective Shadow**, and cross-ratio preservation is its **Projective Fingerprint**. Boundedness alone does not select this branch.

The broader **Empirical Grammar** then asks whether an empirical system is entitled to that geometry. Scalar closure is tested rather than assumed; deterministic point geometry is licensed only after a **Point-Map Gate**; projective rank is complexity-controlled; and when matched observable states produce different futures, the **False-Noun Criterion** and **Failure Tomography** direct state enlargement. A reproducible synthetic unit test demonstrates quotient recovery, projective identification, controlled hidden-state failure, robustness under noise and misspecification, and held-out recovery. A quantitatively preregisterable HepG2/Grx1-roGFP2 protocol supplies a prospective natural-system test.

The larger methodological implication is that scientific disagreement is well posed only after the observational lens, intervention jurisdiction, predictive state, horizon, and future records have been made common. The paper does not claim that all observation is projective or that consciousness generates physical law. It claims something more elementary: **we do not stand outside nature when we describe it. A scientific state is a compression of history; a geometry records what a lens preserves; and failed prediction is evidence that the compression erased a distinction the future can still reveal.**

Claim architecture and evidential status

Layer	Claim	Status	What can defeat it
Predictive closure	Histories with identical future laws define the minimal predictive state relative to declared access	established construction / constitutive	finite-data reconstruction may fail to recover the population quotient
Observation geometry	Scale-blind two-channel homogeneous linear dynamics induce a Möbius law; cross-ratio gives the converse fingerprint	proved / classical equivalence assembled here	mathematical counterexample to the stated hypotheses or proof
Empirical Grammar	State, closure, point geometry, projective rank and failure are tested in that order	methodological synthesis + decision architecture	prospective failure to identify or recover predictive state under declared gates
Synthetic study	The procedure recovers, breaks and repairs the intended architecture under controlled stress	demonstrated in silico	reproducibility failure or adverse stress tests
HepG2 redox	A fast projective jurisdiction may fail when slower NADPH/NRF2 capacity becomes predictive	prospective hypothesis	any preregistered gate failure
Broader programme	Scientific disputes and first-/third-person observation can be typed by lens, jurisdiction, state and future law	programmatic implication	failure to produce useful discriminating tests across natural disputes

Science from Within Nature

Every scientific measurement is made through a physical interface. The observer is not an external spectator looking down on a completed catalogue of states. It is a subsystem of the world with finite instruments, interventions, records, resolution and memory. Science therefore begins neither with a metaphysical state space nor with a coordinate system. It begins with **what can be done, what can be recorded, and which distinctions the future can still reveal**.

This does not make truth subjective. It makes the conditions of access explicit. The raw repertoire is itself theory-laden: instruments are built using prior physics, chemistry and engineering. The first-principles claim is relative to that physical access layer. Once the interface is declared, however, the latent state geometry need not be smuggled in with it.

A scientific state can then be read as a promise. To place two histories in the same state is to claim that every distinction between them has become irrelevant to every declared future test. If the future later separates them, the promise was false and the state was too small. In this sense a state is **surviving predictive history**: a compression of the past justified only by closure of the future.

Geometry is downstream of that compression. One particularly important lens discards **common scale**. Suppose a latent description contains two nonzero homogeneous channels

$$z = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix},$$

but the observer cannot distinguish z from λz , with $\lambda \neq 0$. Operationally, the accessible object is then a ray $[z_1 : z_2]$, or a coordinate such as

$$q = \frac{z_1}{z_2}, \quad x = \frac{z_1 - z_2}{z_1 + z_2}.$$

If those hidden channels transform linearly,

$$z' = Mz, \quad M = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

then the visible ratio obeys

$$q' = \frac{az_1 + bz_2}{cz_1 + dz_2} = \frac{aq + b}{cq + d}.$$

Nothing nonlinear has been inserted into the source. The fractional-linear law is what remains after one degree of freedom - common scale - has become inaccessible. This is the **Projective Shadow**:

linear homogeneous source $\xrightarrow{\text{loss of common scale}}$ projective empirical state.

The exact boundary matters. Finite observation by itself does not imply projective geometry. Arbitrary compression of a linear system is not generally projective. Theorem 1 applies when the observation map is homogeneous or scale-blind in the stated sense. The power of the claim comes from this narrowness: when those hypotheses are earned, the geometry is not chosen.

Corollary 1: The Projective Equivalence Principle

Theorem 1 immediately explains why the same Möbius law can recur in sciences that share no physical substrate.

Corollary 1 (Projective Equivalence Principle). If two empirical domains independently establish observables that are scale-blind coordinates of homogeneous linear processes, then their induced transformation families belong to the same projective class: in one-dimensional charts they are fractional-linear and preserve cross-ratios. Their physical meanings may be entirely different; the invariant grammar of the observer-world interface is the same.

This is a unification of representation, not ontology:

same observational constraint $\Rightarrow$ same induced mathematics, while same mathematics $\nRightarrow$ same physics.

The cross-ratio is the **Projective Fingerprint** that permits this equivalence class to be tested from the observable side.

Calibration cases: the corollary already realized

Known systems provide calibration cases for Corollary 1; they are not counted as discoveries of this framework.

Jurisdiction	Independently known latent object	Observable projective coordinate	What the framework recovers
Collinear relativity	light-cone components	$\beta = v/c$	rapidity and Möbius velocity addition [8,9]
Celestial directions	two-spinor	stereographic sky coordinate	$SL(2, \mathbb{C})$ Möbius action [13]

Jurisdiction	Independently known latent object	Observable projective coordinate	What the framework recovers
Lossless multilayers	forward/backward amplitudes	reflection coefficient	$SU(1, 1)$ disk action [10-12]
Gaussian optics	ABCD linear pair	beam parameter q	fractional-linear propagation [14]
Microwave networks	two-port wave amplitudes	Γ , impedance	Smith-chart Möbius maps [15,16]
Binary inference	two positive hypothesis weights	posterior odds	additive log-odds
Two-allele fixed selection	two type abundances	allele-frequency odds	additive log relative fitness [17]

The common explanation is not that these domains share a physical substrate. Each independently supplies linear homogeneous channels, and normalization removes common scale. The projective law is then the same representation-theoretic consequence. The catalogue therefore validates the **typing of the mechanism**, while the synthetic and prospective biological tests are where the framework itself takes discovery risk.

A particularly useful comparison is binary inference versus fixed-fitness selection. Both obey

$$\text{new odds} = \text{multiplicative factor} \times \text{old odds},$$

so their log-odds are additive even though one factor is a likelihood ratio and the other a relative fitness. This is a clean example of shared grammar without shared ontology.

Projective Equivalence Principle - Rosetta Stone form. Wherever the same representational constraint is independently established, the recurrence of the Möbius law is not an unexplained coincidence: scale-blind observation of linear homogeneous state necessarily casts a Projective Shadow. The cross-ratio is its operational fingerprint. Failure Tomography is the protocol for discovering predictive structure that lies beyond that shadow.

Projectivity is not boundedness

The observer-centered theorem also identifies an important limit. A bounded observable does not, merely by being bounded, inherit Möbius composition or hyperbolic geometry.

Let

$$\phi : (-1, 1) \rightarrow \mathbb{R}$$

be any smooth increasing bijection and define

$$T_t(x) = \phi^{-1}(\phi(x) + t).$$

Then T_t is a smooth, reversible, transitive one-parameter action on the bounded interval. For generic ϕ , however, it is not Möbius. Boundedness therefore determines neither the projective group nor the coordinate $\text{artanh } x$.

The missing condition is structural. The Projective Shadow appears when the empirical variable is a projective coordinate of an appropriate homogeneous linear state - or, equivalently on the projective line, when its transformation family exhibits the Projective Fingerprint.

$$\text{bounded} \neq \text{projective}.$$

Only after projectivity is earned can hyperbolic subclasses be considered.

From observer theorem to empirical grammar

The universal kernel and the exact branch. The Projective Shadow is not the universal layer of the framework. ****Predictive closure is.**** Every application first asks which histories and interventions remain distinguishable by future law. Projective geometry enters only after a sufficient state has been earned, a point map is licensed, and scale-blind homogeneous linear structure survives testing.

The Observation Projection Theorem begins with a state on which transformations act. Empirical science does not generally begin with such a state already identified. Before asking whether a state transforms projectively, one must determine what the state is, what interventions are effectively distinct, whether one scalar closes, and whether the next predictive state is sufficiently deterministic for a point transformation to be meaningful.

That is why the rest of this paper has the form of a grammar rather than a universal projective model. The theorem supplies the observer geometry for one sharply defined branch. The grammar determines when an empirical system is entitled to enter that branch and what to do when it does not.

The order is

physical access $\rightarrow$ predictive distinction $\rightarrow$ state $\rightarrow$ closure $\rightarrow$ Point-Map Gate $\rightarrow$ geometry.

When a proposed compression fails, the future itself identifies the problem:

same recorded present + different future $\Rightarrow$ the representation forgot something predictive.

That is the **False-Noun Criterion**. Under the appropriate smoothness conditions, reversal defects probe Lie-bracket directions; at finite amplitude, cycle circulation tests global scalar integrability. Together these constitute **Failure Tomography**. No inferred variable is accepted merely because it repairs a fit: the enlarged state must improve prediction on data that were not used to construct it.

The workflow is therefore not the primary discovery. It is the operational consequence of the observer theorem:

Earn state. Test the lens. Use failure to reconstruct. Let held-out prediction decide.

The full twelve-step implementation and decision diagram are collected in **Appendix A**, so that the theorem remains visually primary in the main argument.

What the Grammar Adds Beyond Causal States and PSRs

The predictive-state layer is established work. Causal states, epsilon-transducers, and predictive-state representations construct future-sufficient state from observations and controlled action-observation sequences [1-3]. CSSR and spectral PSR methods provide concrete learning procedures under explicit assumptions [24,25,26]. The **Empirical Grammar is not a replacement estimator**. It is a decision layer placed around those estimators.

Question	Causal states / epsilon-transducers / PSRs	Additional constraint supplied here
What state is sufficient for future prediction?	Core problem; established	Uses the learned state as the starting object, not as the conclusion

Question	Causal states / epsilon-transducers / PSRs	Additional constraint supplied here
Which raw actions are empirically distinct?	Inputs are ordinarily supplied as action labels	Quotients interventions by future-law indistinguishability to obtain an effective jurisdiction
When may a deterministic geometry be fitted?	Not a canonical PSR step	Point-Map Gate requires the next predictive state to be sufficiently concentrated around a point map
What observational condition selects projective geometry?	Not supplied by predictive-state reconstruction	Observation Projection Theorem: scale-blind access to homogeneous linear state forces the projective branch
How is the branch tested from observables?	No projective invariant is built into the standard construction	Projective Fingerprint: held-out Möbius prediction plus cross-ratio preservation
What happens when one scalar is inadequate?	State splitting/model refinement is possible	False-Noun Criterion plus local reversal and global cycle Failure Tomography
How is geometric dimension prevented from inflating indefinitely?	General model selection applies	Projective channel count is pre-bounded and complexity-penalized before the final test

The distinction is therefore precise. Predictive-state methods answer **what must be remembered for prediction**. The additional grammar asks **what operations are effectively distinct, whether a point geometry is licensed, which geometry is forced by the observer's lens, and what structured experiment should follow when a proposed representation fails**. The projective mathematics itself is classical; the contribution is the observer-world typing and the sequence of empirical gates that makes that branch falsifiable.

The forward algebra of Theorem 1 remains intentionally elementary. Scientific content resides in identifying scale-blindness as the observational condition that forces projective geometry and in turning cross-ratio preservation into a held-out test rather than a coordinate choice.

Where State Begins: Actions, Records, and the Birth of the Quotient

Before jurisdiction: the raw operational repertoire

The main apparent circularity in an operational reconstruction is immediate. Predictive state is defined relative to a family of admissible future tests, but how can those tests be specified without already knowing the world's state space and causal structure?

The answer is to distinguish an **operational repertoire** from an **effective jurisdiction**.

Let $\mathcal{U}_0$ be a raw set of intervention labels corresponding to procedures the apparatus can physically execute, and let $\mathcal{Y}_0$ be the record alphabet or record sigma-algebra produced by the apparatus. The labels need not yet possess a metric, topology or state-space interpretation. They may be switch settings, pulse amplitudes, waveform files, reagent additions, control paths or continuous actuator trajectories.

This repertoire is **not theory-free**. An apparatus exists because prior physics and engineering made some operations executable and some records measurable; even choosing a pulse amplitude rather than a waveform is a modeling decision. The claim here is therefore relative, not foundational in the sense of escaping all prior theory: once a physical access layer has been built, predictive state can be reconstructed without additionally assuming the state geometry that the reconstruction is meant to test. New instruments change that access layer and can refine the resulting state.

A finite history is written

$$h_t = (u_1, y_1; \dots; u_t, y_t),$$

or, in continuous time, by the corresponding control and record paths. An empirical model assigns conditional laws

$$P(A \mid h, u_{1:k})$$

for future record events A under future executable sequences $u_{1:k} \in \mathcal{U}_0^k$.

No state manifold appears in these definitions.

History equivalence

For a fixed horizon family and raw repertoire, define

$$h \sim h'$$

when

$$P(A \mid h, u_{1:k}) = P(A \mid h', u_{1:k})$$

for every admissible future sequence and every future record event. The predictive state is

$$S(h) = [h]_{\sim}.$$

This is the controlled predictive-state or causal-state construction [1-3].

Intervention equivalence

Two raw interventions $u, v \in \mathcal{U}_0$ are predictively equivalent, written

$$u \approx v,$$

if replacing one by the other never changes the law of the complete future record, for any history and any later continuation. Explicitly, for every history h , continuation $w_{2:k}$, and future record event A ,

$$P(A \mid h, u, w_{2:k}) = P(A \mid h, v, w_{2:k}).$$

The **effective jurisdiction** is the quotient

$$\mathcal{J}^* = \mathcal{U}_0 / \approx.$$

Thus physically different buttons that are predictively indistinguishable occupy the same operational action class.

Minimal joint predictive reduction

Relative to the declared access layer and horizon, the history and intervention quotients are the **coarsest** representations preserving all future-law distinctions. If a history representation R is sufficient,

$$R(h) = R(h') \Rightarrow h \sim h',$$

then its fibers refine predictive equivalence and there is a unique map on $\text{im } R$ such that

$$S = f \circ R.$$

The same factorization holds for any sufficient intervention representation and $\mathcal{J}^*$. The direction matters: a richer representation may retain distinctions prediction does not need, but every sufficient

representation can be compressed to the predictive quotient. The construction is canonical **relative to the physical access layer**, which remains theory-laden and may change when new instruments are built.

Adaptive policies do not require a second state definition

For measurable non-anticipating policies, open-loop predictive equivalence is already sufficient. If two histories induce identical future-record laws for every fixed action sequence through horizon T , then induction over the record prefix gives identical laws under every policy whose next action depends only on the observable past. Fixed sequences are degenerate policies, so the converse is immediate. Adaptive experimentation therefore changes **which distinctions are efficiently discovered**, not the definition of predictive state.

Jurisdiction refinement

New instruments enlarge the executable repertoire. If $\mathcal{U}_1 \subseteq \mathcal{U}_2$, equivalence under the richer repertoire implies equivalence under the poorer one, giving a canonical surjection

$$S_{\mathcal{U}_2} \twoheadrightarrow S_{\mathcal{U}_1}.$$

Thus scientific state spaces form a refinement system rather than a fixed ontology: new actions or records can split an old predictive class without making it wrong in its former jurisdiction.

Learning predictive state and jurisdiction from finite data

Exact predictive equivalence is a population object. With finite data, choose a distance D between conditional future laws and define finite-horizon predictive and intervention pseudometrics. Because ϵ -closeness is not transitive, finite-resolution state is a **model-selection problem**, not an exact quotient produced by thresholding noisy pairwise distances. This connects directly to probabilistic bisimulation metrics, CSSR, and spectral PSR learning [21,22,24,25,26].

Algorithm 1. Confidence-aware predictive reconstruction

1. Predeclare the horizon, probe policies, observation resolution, estimator, and reconstruction/validation/test splits.
2. Estimate conditional future laws for sampled histories and probes, with uncertainty.
3. Cluster by a diameter-controlled rule such as complete linkage rather than single-link chaining.
4. Validate future-law homogeneity on independent data and split rejected clusters.
5. Apply the same procedure to intervention-conditioned laws to estimate $\mathcal{J}^*$.
6. Select new probes for expected information gain, then freeze the architecture before the final held-out test.

Under a finite separated-state model, uniform consistency of the predictive-law estimates and a threshold lying strictly between within-state estimation error and the minimum between-state separation are sufficient for complete-link recovery of the true partition with probability tending to one. This is a standard margin argument, not a universal sample-complexity theorem. Continuous-state, weakly separated, long-memory, and adaptively collected processes require the richer causal-state/PSR/statistical-learning literature. The grammar supplies the **decision layer** around those estimators.

State adequacy and the false-noun criterion

Scientific practice rarely manipulates the predictive quotient directly. It proposes variables

$$L : \mathcal{H} \rightarrow \mathcal{X},$$

such as dose, temperature, concentration, velocity, genotype frequency, disease stage or a vector of biomarkers.

State adequacy criterion

A proposed variable L is sufficient for the stated jurisdiction exactly when

$$L(h) = L(h') \implies h \sim h'.$$

Equivalently, there exists a map f such that

$$S = f \circ L.$$

If equal values of L lead to different future laws, the representation has compressed too aggressively.

the measurement has forgotten a distinction the future can still reveal.

The remedy is structural, not rhetorical: enlarge the representation to (L, Z) , or to a richer object, until future prediction closes.

The statement

state = surviving predictive history

is therefore not an ontological claim about what reality is made of. It is the canonical answer to a modeling question: which distinctions from the past remain necessary for the declared future tests?

When One Number Is Enough: The Closure Test

Sequential composition and the closure trichotomy

Suppose a proposed scalar summary $L(h) \in I \subseteq \mathbb{R}$ is intended to summarize sequential history. Ask whether there exists

$$F : I \times I \rightarrow I$$

such that

$$L(h_1; h_2) = F(L(h_1), L(h_2)).$$

If such an F exists, associativity of history concatenation forces

$$F(F(x, y), z) = F(x, F(y, z)).$$

Closure classification

Relative to the regular ordered Aczél branch, a proposed scalar sequential summary has three possible fates.

Case I: no scalar closure. No single-valued F exists. If equal scalar values support different future laws, the scalar is not a sufficient state.

Case II: lawful scalar closure outside the regular additive class. A single-valued associative law exists but cancellativity, strict monotonicity or another regularity hypothesis fails. Idempotent, absorbing and max/min-like semigroups are examples. Such laws can erase distinctions irreversibly while remaining lawful.

Case III: regular ordered scalar closure. If F is continuous and strictly monotone in each argument on an interval, an additive natural coordinate exists.

Strict monotonicity is not cosmetic. True plateaus, absorbing endpoints, hysteresis, order-dependent loops, or other failures of cancellation do not belong to Case III merely because a smooth sigmoid can be fitted. They either belong to Case II or indicate that the scalar itself is not a closed state. A bounded saturating curve on an open interval may still be strictly monotone; an actual plateau or hysteretic branch is a different algebraic object.

The classification concerns the fate of a proposed scalar sequential summary. It is not a classification of all physical theories.

Aczél representation for regular closure (established)

Under the Case III hypotheses there exists a continuous strictly monotone ϕ such that

$$\phi(F(x, y)) = \phi(x) + \phi(y).$$

Equivalently,

$$F(x, y) = \phi^{-1}(\phi(x) + \phi(y)).$$

Thus additivity can be the natural coordinate representation of lawful composition rather than a primitive assumption.

Boundedness does not select projectivity

A bounded interval does not uniquely select

$$\phi(x) = \operatorname{artanh} x.$$

Boundedness no-go: reversible bounded closure does not force Möbius dynamics

Let $\psi : (-1, 1) \rightarrow \mathbb{R}$ be any smooth increasing bijection and define

$$T_t(x) = \psi^{-1}(\psi(x) + t).$$

Then $\{T_t\}_{t \in \mathbb{R}}$ is a smooth, reversible, transitive one-parameter group of order-preserving transformations of the bounded interval. For generic ψ , these maps are not fractional-linear.

For example,

$$\psi(x) = \tan \frac{\pi x}{2}$$

gives

$$T_t(x) = \frac{2}{\pi} \arctan\left(\tan \frac{\pi x}{2} + t\right),$$

which remains inside $(-1, 1)$ for every finite t but is not Möbius in x .

Therefore

$$\text{boundedness} + \text{reversible scalar closure} \not\Rightarrow \text{projective geometry.}$$

The empirical question is not whether an observable saturates. It is which invariants its transformation family preserves.

Earning the Right to Use Geometry: The Point-Map Gate

The predictive quotient is generally stochastic. After an action u , the next predictive state may branch because the next record is random. Let

$$K_u(s, B) = P(S_{t+1} \in B \mid S_t = s, u_t = u)$$

be the induced transition kernel on predictive states.

Deterministically controlled predictive systems

A jurisdiction is **deterministically controlled at the chosen sampling event** if for every effective action u and state s there exists a point map f_u such that

$$K_u(s, \cdot) = \delta_{f_u(s)}.$$

Only under this condition, or a declared approximation to it, is it legitimate to treat an intervention as a transformation of points and test Möbius/cross-ratio geometry on $s \mapsto f_u(s)$.

The condition is empirical. Repeated preparations of the same reconstructed state under the same action should yield next-state dispersion compatible with the declared observation/reconstruction error. If a non-degenerate transition kernel remains after accounting for that uncertainty, a deterministic projective point map is rejected. One must then model K_u directly, or find a richer predictive state in which the transition becomes deterministic; applying a Möbius fit to a conditional mean is not a substitute for this gate.

No universal numerical Point-Map threshold is implied by the theory. In any application the gate is a **domain-specific equivalence test**: its margin must be fixed before confirmatory testing from measurement resolution, reconstruction uncertainty, and the scientific scale at which a point map would count as adequate. The value 0.020 used later is a frozen design margin for the synthetic and HepG2 tests, not a constant of the grammar.

This distinction separates two questions that are often conflated:

predictive sufficiency is compatible with stochasticity; point geometry is not automatic.

The Projective Shadow: Proof, Fingerprint, and Higher-Rank Generalization

Proof and generalization of Theorem 1

Theorem 1 was stated at the outset because it is the mathematical centre of the paper. This section gives the explicit derivation, the bounded-contrast coordinate, the converse cross-ratio characterization, the higher-rank extension, and the continuous Riccati form.

The forward algebra is elementary: it is division by a denominator. The scientific content is not the algebraic difficulty. It is the identification of **scale-blindness at the observer-world interface** as the condition that makes projective geometry unavoidable for homogeneous linear dynamics, together with the converse use of cross-ratio preservation as an observable fingerprint. The result is therefore not a claim that fractional-linear maps are new mathematics; it is a claim about when that classical mathematics is forced by the structure of measurement and how its failure can be used diagnostically.

A finite observer need not observe ratios, and arbitrary compression of a linear system is not generally projective. The projective mechanism requires a specific loss of information: common scale is unobservable, so empirical state is a ray or ratio of homogeneous channels. Under that condition the geometry is forced.

Let the latent state be two-dimensional:

$$z = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}, \quad z' = Mz, \quad M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \det M \neq 0.$$

Define the observable projective coordinate

$$q = \frac{z_1}{z_2}.$$

Forward direction: linear homogeneous lift to Möbius law

Where the denominator is nonzero,

$$q' = \frac{aq + b}{cq + d}.$$

Proof

Since

$$z'_1 = az_1 + bz_2, \quad z'_2 = cz_1 + dz_2,$$

division by z_2 gives the result. $\square$

The mathematical statement is elementary. Its scientific content is precise: ratio observation converts linear latent dynamics into fractional-linear empirical dynamics.

Positive channels and bounded contrasts

For positive channels $z_+, z_- > 0$, define

$$x = \frac{z_+ - z_-}{z_+ + z_-}.$$

Then

$$q = \frac{z_+}{z_-} = \frac{1+x}{1-x}$$

and

$$r = \frac{1}{2} \ln q = \operatorname{artanh} x.$$

If the latent action is diagonal,

$$z'_+ = e^\rho z_+, \quad z'_- = e^{-\rho} z_-,$$

then

$$r' = r + \rho$$

and

$$x' = \tanh(\operatorname{artanh} x + \rho) = \frac{x + \tanh \rho}{1 + x \tanh \rho}.$$

Artanh is therefore not selected by boundedness alone. It is selected when the bounded observable is a normalized contrast of two positive channels whose log-ratio is translated by the intervention.

The Projective Fingerprint: Cross-Ratio and the Converse Lift

The forward direction becomes an empirical discovery principle only if the implication can also be reversed at the level of observable transformations.

For four distinct points on the projective line define the cross-ratio

$$[a, b; c, d] = \frac{(a-c)(b-d)}{(a-d)(b-c)}.$$

Converse direction: the Projective Fingerprint

Let T be a continuous injective transformation of a real interval into the real projective line. The following are equivalent on its domain:

1. T preserves the cross-ratio of every admissible quadruple.
2. T is fractional-linear,

$$T(q) = \frac{aq + b}{cq + d}, \quad ad - bc \neq 0.$$

3. T is the projective action of an invertible two-dimensional linear map on homogeneous coordinates, unique up to multiplication of the matrix by a nonzero scalar.

Proof

The implication $(3) \Rightarrow (2)$ is the two-channel projectivization identity, and direct substitution gives $(2) \Rightarrow (1)$. For $(1) \Rightarrow (2)$, choose three distinct anchor points q_1, q_2, q_3 . There is a unique Möbius transformation M taking them to $T(q_1), T(q_2), T(q_3)$. For any fourth point q , both $T(q)$ and $M(q)$ have the same cross-ratio with the three image anchors. The cross-ratio with three fixed distinct anchors is injective in the fourth coordinate, hence $T(q) = M(q)$. Finally, a Möbius map is represented by an invertible 2×2 matrix unique up to nonzero scale. $\square$

This theorem supplies the operational inversion missing from a purely sufficient linear-lift argument:

$\begin{aligned} \text{cross-ratio preservation} &\iff \text{fractional-linear empirical law} \\ &\iff \text{abstract 2D linear homogeneous lift.} \end{aligned}$

The theorem does **not** identify the physical meaning of the homogeneous channels. That remains a domain question. But it means a successful cross-ratio test is not merely a convenient curve fit: it is exactly the condition for a two-dimensional projective linear representation of the transformation family.

Beyond two channels: projective spaces and Grassmannians

The two-channel theorem is the rank-one case of a wider construction.

n-channel projectivization

Let $z \in \mathbb{F}^n \setminus \{0\}$, where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, and let

$$z' = Az, \quad A \in GL(n, \mathbb{F}).$$

Observation of the ray $[z]$ gives a state in projective space

$$[z] \in \mathbb{P}^{n-1}(\mathbb{F}).$$

In the affine chart $z_n \neq 0$, with coordinates

$$x_i = \frac{z_i}{z_n}, \quad i = 1, \dots, n-1,$$

the induced transformation is

$x'_i = \frac{a_{in} + \sum_{j=1}^{n-1} a_{ij}x_j}{a_{nn} + \sum_{j=1}^{n-1} a_{nj}x_j}, \quad i = 1, \dots, n-1.$
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Thus an n -component linear latent state produces a multidimensional projective empirical geometry rather than a privileged scalar Möbius law.

Subspace-valued state and the Grassmannian

Sometimes the observable empirical object is not a ray but a k -dimensional latent subspace. Such states live on a Grassmannian $\text{Gr}(k, n)$. In a local graph chart, write the subspace as

$$\{(x, Xx) : x \in \mathbb{F}^k\},$$

and partition an invertible linear map as

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}.$$

Where $A + BX$ is invertible, the induced chart update is

$$X' = (C + DX)(A + BX)^{-1}.$$

This is the matrix analogue of a Möbius transformation. Continuous linear flows on the ambient space induce matrix Riccati equations on these charts; Grassmannian formulations of Riccati dynamics are established [23].

The hierarchy is therefore

$$\begin{aligned} 2 \text{ channels} &\rightarrow \mathbb{P}^1 \rightarrow \text{scalar Möbius/Riccati}, \\ n \text{ channels} &\rightarrow \mathbb{P}^{n-1} \rightarrow \text{multidimensional projective flow}, \\ k\text{-subspaces in } n \text{ channels} &\rightarrow \text{Gr}(k, n) \rightarrow \text{matrix fractional-linear/Riccati flow}. \end{aligned}$$

This is the natural route beyond rank one. It also changes the biological question: failure of a two-channel fit need not imply absence of projective structure; it may indicate that the empirical state is the shadow of a higher-dimensional linear latent system.

Projective dimension is a model-selection variable

Allowing arbitrary latent dimension after every failure would make “projective somewhere” scientifically empty. The candidate homogeneous dimension must therefore be treated exactly like any other model-complexity parameter.

For $d = 2, 3, \dots, d_{\max}$, let a candidate model consist of an empirical embedding

$$\eta_d : S \rightarrow \mathbb{P}^{d-1}$$

and action matrices $M_{u,d}$ acting projectively on that embedding. Fit each candidate on reconstruction data and score it on held-out future laws with a proper predictive score. Select the smallest d whose held-out performance is statistically indistinguishable from the best candidate, or equivalently use a predeclared MDL/BIC-type complexity penalty when its likelihood assumptions are appropriate.

Two safeguards are essential.

1. **Predeclare $d_{\max}$.** Failure of every candidate up to $d_{\max}$ rejects the finite-rank projective family within the searched jurisdiction rather than licensing unlimited rank inflation.
2. **Demand identifiable coordinates.** A scalar record does not by itself identify an arbitrary $\mathbb{P}^{d-1}$ state. Increasing d requires additional predictive coordinates - measured variables, delay coordinates, or other independently validated features - sufficient to distinguish the higher-dimensional rays.

The cross-ratio test is therefore the especially sharp $d = 2$ case, not a promise that any failed scalar law can be rescued by adding enough projective channels. In dimensions $d > 2$, the invariant structure is richer - incidence relations, minors and higher projective invariants replace a single scalar cross-ratio test - and this paper does not claim an equally simple universal finite-sample fingerprint.

The operationally complete branch here is therefore $d = 2$. Higher rank is tested only when additional predictive coordinates are independently identifiable and the predeclared complexity rule supports them.

Continuous projectivization and Riccati dynamics

Let

$$\dot{z} = A(t)z, \quad A(t) = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

For $q = z_1/z_2$, direct differentiation gives

$$\dot{q} = b + (a - d)q - cq^2,$$

up to the numerator/denominator convention.

Linear flow and Riccati projection (established)

The projective coordinate of a two-dimensional linear system satisfies a Riccati equation. Conversely, a scalar Riccati equation can be locally homogenized to a two-dimensional linear system. Finite-time projective maps are Möbius transformations [6,23].

Thus the projective-lift equivalence has both discrete and continuous forms.

Geometry is selected after projectivity

Projectivity is the parent structure. Hyperbolicity is one branch.

- $SL(2, \mathbb{C})$ acting on $\mathbb{CP}^1$ yields complex projective/Möbius geometry on the Riemann sphere.
- Compact unitary subgroups produce rotational or spherical structures.
- $SU(1, 1)$ preserves the unit disk and an indefinite Hermitian form, giving the Poincare disk.
- The real fixed-ray subgroup acting on $(-1, 1)$ yields the artanh/rapidity coordinate.

The correct dependency is

linear lift $\rightarrow$ Projective Shadow $\rightarrow$ invariant subgroup/domain $\rightarrow$ geometry.

Regular scalar branches

A regular Aczél scalar branch can therefore be separated experimentally into

III_N : regular scalar closure without cross-ratio preservation, III_P : regular projective closure with cross-ratio preservation.

Failure of cross-ratio preservation does not imply hidden state. A scalar may close perfectly well in a non-projective Aczél coordinate.

Small-excursion linearity

For the fixed-ray hyperbolic interval,

$$\text{artanh } x = x + \frac{x^3}{3} + O(x^5),$$

so sufficiently small excursions satisfy $r \approx x$. Local linear response is therefore weak evidence about global geometry; a tangent chart can hide finite boundaries and nontrivial global composition.

Synthetic Validation: End-to-End Reconstruction

End-to-end synthetic reconstruction

A decision procedure should perform its own job before asking a biological experiment to trust it. Figure 2 is therefore a **reproducible synthetic unit test** in which the latent architecture is known to the simulator but withheld from the reconstruction. The workflow is required to recover a quotient, identify a projective law out of sample, fail when a hidden channel is introduced, diagnose that failure, and restore prediction only after the missing coordinate is admitted.

Stage A: recover state and effective action classes

Six distinct latent rays were each instantiated at three different common scales, producing 18 raw histories. Future-law features were estimated under three probe actions. Because common scale is predictively irrelevant in the two-channel system, diameter-controlled clustering recovered exactly six predictive states from the 18 histories. The largest within-state feature distance was 0.0016, while the smallest between-state distance was 0.246.

The raw action repertoire also contained two physically distinct matrices, A and $A^* = 2.35A$. They produce identical projective transformations because the common matrix scale cancels. Their estimated intervention distance was 0.0006, compared with at least 0.056 for every non-equivalent action pair. Thus the same data reduction recovered both the history quotient and an intervention quotient.

Stage B: identify the projective fingerprint

A three-parameter Möbius law was fitted on independent two-channel transitions and frozen before testing. On held-out states it achieved RMSE

$$0.000272.$$

Cross-ratio preservation was tested separately on noisy replicate estimates over unused quadruples. The median absolute defect

$$\left| \log \left| \frac{\text{CR}(Tq_1, Tq_2; Tq_3, Tq_4)}{\text{CR}(q_1, q_2; q_3, q_4)} \right| \right|$$

was 0.000434, with 95th percentile 0.003566. This is the operational fingerprint; the Möbius fit is then the corresponding predictive representation.

Stage C: Same Shadow, Different Futures

A third positive channel z_3 was introduced and coupled into the next-step evolution of z_1, z_2 , while the observer retained only

$$x = \frac{z_1 - z_2}{z_1 + z_2}.$$

Histories with the same observed x now had different futures ordered by the hidden ratio $\rho = z_3/(z_1 + z_2)$. The best scalar Möbius model deteriorated to held-out RMSE

$$0.0416.$$

When ρ was admitted as an additional predictive coordinate, the appropriate two-coordinate fractional-linear model reduced held-out RMSE to

$$0.000271,$$

a factor of 153.4. A complexity-penalized comparison strongly favored the enlarged model in this synthetic setting ($\Delta\text{BIC} \approx 4195$ relative to the scalar model).

Stage D: The Missing Dimension Restores the Law

Two small smooth interventions generated by noncommuting 3×3 matrices were then reversed. Across a grid of amplitudes, the observed difference obeyed

$$\Delta x \approx \kappa ab,$$

with $R^2 = 0.99995$. The fitted coefficient was -0.3771 , close to the differential prediction -0.3856 from the commutator at the initial state; the remaining difference is higher-order in the finite amplitudes.

Synthetic reconstruction: quotient, fingerprint, failure, recovery

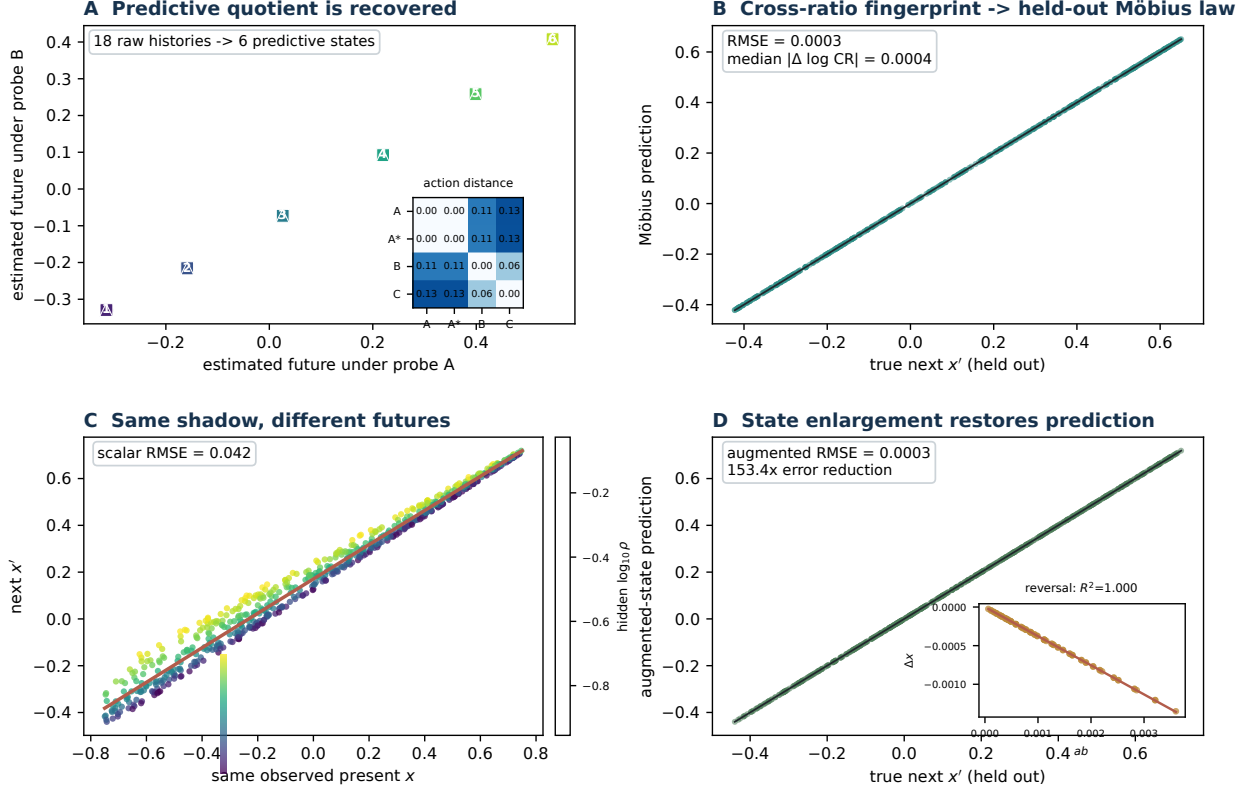


Figure 2: End-to-end synthetic unit test. (A) Eighteen raw histories collapse to six predictive states; the inset shows that two physically different but projectively identical interventions also collapse. (B) The cross-ratio fingerprint and a frozen Möbius law succeed out of sample. (C) Adding an unobserved third channel creates the defining false-state signature: the same present scalar supports different futures. (D) Adding the missing coordinate restores held-out prediction; the inset independently verifies leading reversal-bracket scaling.

Table 3: Synthetic unit-test summary. Each stage is frozen before its held-out score is evaluated.

Stage	What the procedure must recover	Held-out / separation metric
Predictive quotient	18 raw histories $\rightarrow$ 6 predictive states	max within-state distance 0.0016; min between-state 0.246
Action quotient	4 raw interventions $\rightarrow$ effective action classes	$d(A, A^*) = 0.0006$; nearest non-equivalent pair ≥ 0.056
Projective Fingerprint	frozen two-channel Möbius law	RMSE 0.000272; median $ \Delta \log CR = 0.000434$
False-Noun failure	same visible x , hidden third channel	scalar RMSE 0.0416
State enlargement	admit hidden coordinate ρ	RMSE 0.000271; $\Delta BIC \approx 4195$
Failure Tomography	reversal defect vs. ab	$R^2 = 0.99995$; fitted $\kappa = -0.3771$, local prediction - 0.3856

The simulation is not evidence that nature is projective. It is stronger in a different sense: it verifies that the proposed **procedure** can recover a quotient, identify a projective branch, reject an insuf-

ficient scalar, detect a local order defect and recover prediction after state enlargement. The exact script, random seed and generated tables accompany the manuscript.

Stage E: Robustness under noise, misspecification, and uneven sampling

A decision procedure that works only in its generating model is not useful. The frozen simulation was therefore subjected to four additional stress tests. The point-map stress test used 5,000 latent states with a 3,000/2,000 train-test split. The hidden-channel misspecification study used 3,200 states with a 2,000/1,200 split. The reported BIC-style score is the Gaussian residual complexity proxy

$$\text{BIC}^* = n \log(\text{RSS}/n) + k \log n,$$

where n is the held-out sample size and k the fitted parameter count. It is used only as a transparent penalty for otherwise indistinguishable predictive fits. The full script and CSV outputs are supplied in the supplement (`robustness_stress_tests.py`).

Stress test	Result	Interpretation
Latent process noise in the two-channel system	Point-map residual SD rises from 0.0034 at $\sigma = 0$ to 0.0155 at $\sigma = 0.02$, then 0.0304 at $\sigma = 0.04$	With the predeclared gate $s_{\text{resid}} \leq 0.02$, deterministic geometry is accepted through $\sigma = 0.02$ and correctly rejected from $\sigma = 0.04$
Hidden third channel with correlated measurement error ($\rho_e = 0.70$)	scalar Möbius RMSE 0.0417; flexible scalar spline RMSE 0.0419; three-channel projective RMSE 0.00990	A flexible non-projective scalar cannot repair a genuinely multi-valued future; restoring the missing coordinate does
Over-parameterized projective competitor	RMSE 0.00989 versus 0.00990 for the correct three-channel model, but the BIC-style score favors the smaller model by 12.0	Rank/channel inflation does not receive a free pass when held-out fit is statistically indistinguishable
Non-uniform history sampling	36 histories distributed 12/9/6/4/3/2 across the six true states, with only 30-150 probe replicates per history, still recover six clusters with pairwise partition accuracy 1.000	The quotient is not dependent on balanced sampling when predictive separation remains large

The correlated-error stress test also constructs 299 pairs whose observed current coordinates match within 0.02 while their hidden capacities lie in opposite quartiles. Their median absolute future divergence is 0.0916. Thus the False-Noun Criterion survives substantial correlation between measurement error in the visible and hidden coordinates.

A fifth stress test attacks the quotient itself rather than the projective branch. Six true predictive states were compressed toward one another while observation noise and sampling were held fixed (six histories per state, 40 probe replicates per action, observation SD 0.012). Identifiability was summarized by the **separation margin**

$$m_{\text{sep}} = d_{\min}^{\text{between}} - d_{\max}^{\text{within}}.$$

Across 50 Monte Carlo repetitions, a log-ratio span of 0.15 retained a positive margin in 92% of runs; at span 0.12 this fell to 44%, at span 0.10 to 6%, and at span 0.08 to 2%. The intended behavior is therefore not forced recovery. Once between-state separation is no larger than estimation spread, the reconstruction must report ambiguity rather than invent distinct states. This directly exposes the finite-sample identifiability boundary on which any downstream geometry depends.

These tests do not establish universal robustness. They establish something narrower and more useful: the gates fail in the intended direction. Process noise eventually blocks deterministic point geometry; flexible scalar misspecification cannot hide a missing state dimension; a correct enlarged state restores prediction; and a larger projective model is penalized when it adds no predictive value.

When the Shadow Fails: Tomography of Forgotten Dimensions

When does a predictive quotient become a smooth manifold?

History quotients require no topology; Lie brackets do. A smooth calculus is licensed only when the reconstructed family of future laws admits a locally injective finite-dimensional C^r parameterization with constant rank and the usual embedding regularity. Under those standard conditions, the predictive-law image inherits a smooth manifold structure and smoothly parameterized interventions define vector fields on it.

If the predictive image is discrete, singular, stratified, fractal, or discontinuous, the bracket approximation is not used. The finite graph-cycle tests below remain meaningful without inventing a tangent space. Smoothness is therefore an empirical branch condition, not an assumption hidden in the word “history.”

Local failure tomography: reversal and Lie brackets

Assume now that the smooth predictive-realization conditions above apply and that two small interventions are represented by flows generated by vector fields X_A and X_B . Write

$$A_a = e^{aX_A}, \quad B_b = e^{bX_B}.$$

The Baker-Campbell-Hausdorff expansion gives

$$\log(A_a B_b) = aX_A + bX_B + \frac{ab}{2}[X_A, X_B] + O(a^2b + ab^2),$$

while reversing the order changes the sign of the bracket term.

Local reversal-bracket law

For a smooth endpoint $Y : S \rightarrow \mathbb{R}$,

$$Y(A_a B_b s) - Y(B_b A_a s) = ab dY_s([X_A, X_B]) + O(a^2b + ab^2),$$

up to the sign convention used for composition.

A reproducible quadratic small-amplitude reversal contrast therefore measures the endpoint-visible component of a Lie-bracket direction.

The important interpretation is not that every missing variable is a rotation. The bracket can point toward any locally accessible state direction that the original representation failed to preserve.

Iterated commutator sequences probe

$$X_i, \quad [X_i, X_j], \quad [X_i, [X_j, X_k]], \dots$$

and geometric control theory relates the rank of this Lie closure to local accessibility. In a predictive-state experiment this becomes a tomography programme: vary the order and amplitude of interventions and determine how many independent directions the future can distinguish.

Global failure tomography: graph cycles and integrability

Smooth local analysis is not always available. Finite history contrasts provide a complementary test.

Let experimental histories be nodes of a graph and let

$$d_{ij} = -d_{ji}$$

be an antisymmetric pairwise contrast.

Graph cycle-integrability criterion

Let experimental histories be nodes of a connected graph and $d_{ij} = -d_{ji}$ an antisymmetric edge contrast. A scalar node potential V with $d_{ij} = V_j - V_i$ exists **iff** every cycle has zero circulation,

$$\sum_{(i,j) \in C} d_{ij} = 0.$$

Necessity is telescoping; sufficiency follows by defining V_i as a root-to-node path sum, with zero cycle circulation guaranteeing path independence. A triangle is the smallest test; denser graphs distinguish stochastic noise from systematic non-integrability.

Failure Tomography in practice

The word *tomography* is reserved here for a defect that is turned into a predeclared reconstruction loop rather than merely reported as a residual.

1. **Earn smoothness before brackets.** Fit local predictive-law coordinates on training data and require stable finite dimension, local injectivity and approximately constant rank under resampling. If those conditions fail, do not infer tangent directions; use the discrete cycle route.
2. **Run a local reversal scan.** Choose perturbation families A_a, B_b , randomize AB versus BA , and repeat on a shrinking amplitude grid. Regress $\Delta Y/(ab)$ against amplitude. Retain a bracket direction only when its coefficient approaches a stable finite limit and generalizes to held-out initial states.
3. **Rank the defect.** Repeat across independent endpoints or perturbation pairs. Linearly independent held-out response signatures define candidate missing directions; they do not yet name physical variables.
4. **Run a global cycle scan.** Estimate antisymmetric edge contrasts and their covariance on a finite state graph, then test a basis of independent cycle sums. Circulation compatible with zero leaves a scalar node potential admissible; persistent circulation rejects global scalar integrability.
5. **Demand predictive recovery.** Domain variables suggested by the defect are added on training data, frozen, and accepted as state only if they improve held-out future laws after complexity control.

Failure Tomography is therefore a diagnostic-to-reconstruction protocol. A nonzero bracket or cycle defect is evidence that a proposed representation is structurally incomplete; it is not by itself a unique identification of the missing physical coordinate.

A Quantitative Biological Trial: HepG2 Glutathione Redox Across Two Jurisdictions

The prospective biological test is now specified as a registered-report-style protocol rather than a narrative programme. Its purpose is to place three claims independently at risk: (i) a short-window Projective Shadow exists for glutathione redox, (ii) the same scalar becomes insufficient when reductive/adaptive capacity changes, and (iii) measured enlargement of state restores held-out prediction.

System, sensor, and pre-registered windows

The primary system is **HepG2 human hepatoma cells (ATCC HB-8065)** stably expressing cytosolic **Grx1-roGFP2**, a genetically encoded probe that equilibrates rapidly with the glutathione redox couple and supports second-to-minute live-cell measurements [27,28]. HepG2 is chosen because an independent oxidative-stress time series already exists at 0.5, 1, 2, 4, 6, 8, and 24 h using 50 μM H_2O_2 , 100 μM menadione, and 200 μM tert-butyl hydroperoxide (TBH; GEO GSE39291) [29]. Related HepG2 studies show NRF2 stabilization by approximately 60 min and NRF2-target transcription at 4-16 h [32].

Two jurisdictions are fixed **before** outcome analysis:

- **Fast jurisdiction:** -2 to $+20$ min around a 5-min, 50 μM H_2O_2 pulse, with imaging every 15 s. This window is deliberately earlier than the 30-min transcriptomic sampling in GSE39291 and

the approximately 60-min NRF2 stabilization reported under related HepG2 electrophilic stress [29,32]. It is not assumed that transcription is literally zero; only that the predeclared fast state can be tested before slower transcriptional capacity dominates.

- **Adaptive jurisdiction:** 4, 8, and 24 h after 200 μM TBH, matching the published HepG2 oxidative-stress time series [29]. NRF2 dependence is perturbed by 70 nM NRF2 siRNA for 4 h followed by 24 h recovery, a protocol previously used in HepG2 cells [31].

Each biological replicate is an independent culture day/passaging instance. The confirmatory design uses $n = 10$ **biological replicates**, three wells per condition, and at least 100 quality-controlled tracked cells per well for the live-cell analysis. The replicate, not the individual cell, is the inferential unit.

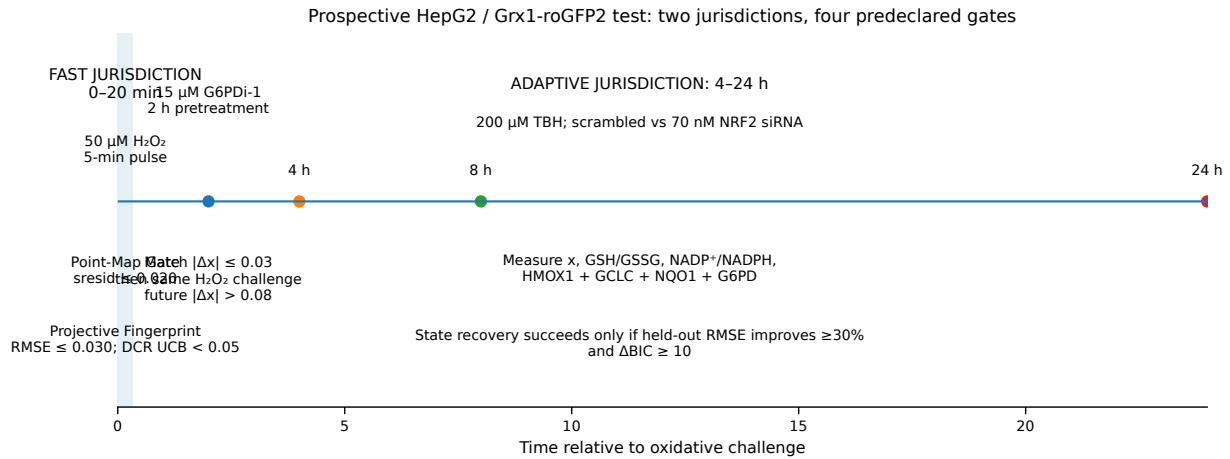


Figure 3: Pre-registered biological test. The fast jurisdiction tests the Point-Map Gate and Projective Fingerprint under a fixed 50 μM H_2O_2 pulse. A G6PDi-1 arm creates a matched-present hidden-capacity challenge. The 4-24 h TBH/NRF2 arm tests whether adding NADPH and NRF2-capacity coordinates restores held-out prediction. Thresholds shown here are fixed before confirmatory acquisition.

Primary fast-window test: the Projective Fingerprint

For the sensor signal, define a calibrated bounded redox coordinate $x \in (-1, 1)$ from the Grx1-roGFP2 oxidation fraction; in parallel destructive wells quantify GSH, GSSG, and $\text{NADP}^+/\text{NADPH}$ by targeted LC-MS or validated enzymatic assays. The physical two-channel hypothesis is written in reducing-equivalent units as

$$R = [\text{GSH}], \quad O = 2[\text{GSSG}], \quad x = \frac{R - O}{R + O}.$$

The Projective Fingerprint is **not** tested by taking four successive time points on one trajectory. It is tested across four distinct initial states under the **same fixed intervention**. Each tracked cell supplies a pair (x_i, x'_i) , with x_i the mean over the final 60 s before the pulse and x'_i the mean over the final 60 s of the 5-min pulse. Seventy percent of cells in each biological replicate calibrate a Möbius map; the remaining 30% are frozen for testing.

Before any projective test, the **Point-Map Gate** must pass. After subtracting the technical sensor variance estimated from repeated baseline frames, the held-out residual standard deviation around the best deterministic map must satisfy

$$s_{\text{point}} \leq 0.020.$$

If this gate fails, the deterministic projective hypothesis is rejected for that jurisdiction and cross-ratio fitting is not used to rescue it.

If the gate passes, two co-primary projective criteria are evaluated:

1. **Held-out Möbius accuracy:** replicate-level held-out RMSE in x must be ≤ 0.030 .

2. **Cross-ratio equivalence:** for quadruples with all pairwise initial-state separations ≥ 0.05 , define

$$D_{\text{CR}} = \text{median} \left| \log \left| \frac{\text{CR}(x'_1, x'_2; x'_3, x'_4)}{\text{CR}(x_1, x_2; x_3, x_4)} \right| \right|.$$

The one-sided 95% upper confidence bound for the mean replicate-level D_{CR} must be **<0.05**. This corresponds to a predeclared approximately 5% multiplicative tolerance in the cross-ratio fingerprint. This 0.05 margin is likewise a **domain-specific preregistered equivalence margin**, not a universal definition of projectivity.

The projective branch is called supported only if the Point-Map Gate and both co-primary criteria pass. Failure of any gate is reported as failure, not repaired by increasing model rank after seeing the final test data.

Hidden-capacity challenge: a pre-specified False-Noun test

The first hidden-capacity perturbation targets NADPH supply with **G6PDi-1**, a rapid reversible cell-active G6PD inhibitor. In HepG2 cells G6PDi-1 reduces 6-phosphogluconate with a cellular IC_{50} of approximately 13 μM ; cellular G6PD/NADPH effects are detectable within minutes in other responsive cells [30]. The protocol therefore uses **15 μM G6PDi-1 for 2 h**, with vehicle controls and parallel target-engagement wells measuring 6-phosphogluconate and $\text{NADP}^+/\text{NADPH}$.

At the end of pretreatment, vehicle and G6PDi-1 cells are matched by nearest-neighbour pairing on the observed redox coordinate with

$$|x_1 - x_2| \leq 0.03.$$

Both groups then receive the same held-out 5-min, 50 μM H_2O_2 challenge. The scalar redox coordinate fails the False-Noun Criterion if matched presents produce a predeclared future separation: the absolute difference in replicate-level mean x at 10 min after challenge must exceed **0.08**, with the 95% confidence interval excluding zero. This test is deliberately directional only in magnitude; the framework predicts separation, not which pretreatment must have the larger signed response.

Adaptive-window challenge: NRF2-dependent capacity

For the 4-24 h jurisdiction, scrambled-control and **70 nM NRF2-siRNA** HepG2 cultures [31] receive **200 μM TBH** and are measured at 4, 8, and 24 h, matching an established HepG2 oxidative-stress exposure schedule [29]. Parallel wells quantify:

- cytosolic Grx1-roGFP2 redox coordinate;
- GSH and GSSG;
- $\text{NADP}^+/\text{NADPH}$;
- HMOX1, GCLC, NQO1, and G6PD transcript abundance as an NRF2-capacity panel;
- viability, with a predeclared requirement of $\geq 90\%$ of vehicle for the trajectory to remain in the intended non-lethal jurisdiction.

The enlarged predictive state is specified before fitting as

$$S_{\text{enlarged}} = (x, \text{NADP}^+/\text{NADPH}, C_{\text{NRF2}}),$$

where C_{NRF2} is the first standardized principal component of the four predeclared NRF2 target transcripts, with the loading vector estimated on training replicates only.

Success requires **leave-one-biological-replicate-out** held-out prediction to improve by at least

$$30\% \text{ in RMSE}$$

relative to the scalar x -only model, together with $\Delta\text{BIC} \geq 10$ in favor of the enlarged state. If the enlarged state does not meet both criteria, the proposed recovery is rejected even if individual coefficients are significant.

Biological validity checks and confounders

The proposed fast two-channel reduction and the later NADPH/NRF2 enlargement are working hypotheses, not unique biochemical decompositions. Three failure modes are explicitly monitored.

- **Sensor and compartment artifacts.** Grx1-roGFP2 calibration drift, photobleaching, segmentation error and compartmental redox heterogeneity are estimated from technical controls. A biological Point-Map pass cannot be manufactured by subtracting unexplained technical variance post hoc.
- **Cell-state heterogeneity.** Cell-cycle state, baseline metabolic capacity and viability may remain predictive even at matched visible redox coordinate. If they explain held-out divergence better than the proposed NADPH/NRF2 variables, the intended enlargement is rejected or expanded under a new preregistration.
- **Damage-induced jurisdiction change.** A perturbation that causes irreversible damage, loss of viability or discontinuous transcriptional state change may leave the proposed smooth jurisdiction rather than merely expose a hidden coordinate. Such observations are typed as jurisdiction failure, not forced into the projective branch.

State recovery also requires **orthogonal validation**. Any recovered capacity coordinate must predict not only future Grx1-roGFP2 trajectories but independent biochemical readouts already specified in the design, including $\text{NADP}^+/\text{NADPH}$ and GSH/GSSG measurements. The same training/held-out split is used for both classes of endpoint.

Power, multiplicity, and preregistration

The confirmatory sample size is fixed at $n = 10$ independent biological replicates. For the fast cross-ratio equivalence test, this gives approximately **82% power** for a one-sided $\alpha = 0.05$ test of the 0.05 margin under the conservative planning assumptions $E[D_{\text{CR}}] = 0.015$ and between-replicate SD 0.04. For the matched-state future-divergence test, $n = 10$ gives approximately **80% power** for a paired standardized effect $d = 1.0$. These assumptions are **planning alternatives, not empirical estimates of the expected HepG2 effect**. The quoted power therefore characterizes the preregistered design conditional on those alternatives. A blinded technical run-in may estimate assay variance but may not change the hypothesis directions, margins, time windows, effect targets, or held-out analysis after confirmatory data are opened.

The gate order is hierarchical, which controls interpretation without post hoc multiplicity fishing: **Point-Map Gate** -> **Projective Fingerprint** -> **False-Noun test** -> **state-enlargement test**. All exclusions, image-quality thresholds, cell tracking rules, cross-ratio quadruple separation criteria, model formulas, random seeds, and held-out folds are to be preregistered before confirmatory acquisition.

Decisive outcomes

The experiment has four clean outcomes:

1. **Fast success, long failure, successful recovery:** the strongest confirmation of the jurisdiction-dependent grammar.
2. **Fast projective failure:** rejects the proposed two-channel linear lift even before adaptive capacity is invoked.
3. **Scalar failure without recovery:** confirms that the redox scalar is insufficient but falsifies the proposed NADPH/NRF2 enlargement.
4. **No scalar failure on extension:** narrows the claim by showing that the redox coordinate remains sufficient over the tested adaptive jurisdiction.

No outcome is labeled a success merely because a more complicated model can be fitted. The biological result is accepted only through predeclared held-out prediction.

This protocol is grounded in established Grx1-roGFP2 redox sensing [27,28], a published HepG2 oxidative-stress time course [29], cell-active G6PD inhibition [30], HepG2 NRF2 knockdown [31], and independent HepG2 NRF2 timing data [32]. It is nevertheless a **prospective test**: none of those studies tested the Observation Projection Theorem or the complete failure-and-recovery sequence proposed here.

Where the Present Result Ends

The strongest claims in this paper are intentionally asymmetric. The predictive quotient is a representation construction relative to stated future laws. The Observation Projection Theorem is exact in its two-channel scale-blind linear branch. The Empirical Grammar is a methodological architecture whose external validity remains to be established prospectively.

Four boundaries matter most. First, **finite-sample identifiability** can fail before geometry is reached; the weak-separation stress test makes that boundary explicit. Second, **jurisdiction is physical and revisable**: new instruments or longer horizons can refine the state. Third, the simple cross-ratio fingerprint is special to $\mathbb{P}^1$; higher-dimensional projective identification requires richer observables and invariants. Fourth, the glutathione programme is a prospective test. No synthetic success substitutes for a natural system surviving predeclared gates.

These limits are not retreats from the programme. They are the points at which the programme can be wrong.

Scientific Disagreement After the Lens Is Declared

A consequence of predictive closure is a simple criterion for when two scientific descriptions genuinely disagree empirically. Let two models T_A, T_B be expressed on a common observational jurisdiction $\mathcal{J}$, common predictive state S , common record space and horizon. Define

$$T_A \equiv_{\mathcal{J}, H} T_B \iff P_A(Y_{1:k} \mid s, u_{1:k}) = P_B(Y_{1:k} \mid s, u_{1:k})$$

for every admissible s , intervention sequence and $k \leq H$. If this equality holds, the two descriptions do not make an empirical disagreement inside that jurisdiction, however different their coordinates or interpretation may be. If it fails, a discriminating future experiment exists in principle within the declared access layer.

This suggests a discipline for scientific controversy. Before deciding which camp is wrong, ask whether the apparent conflict is instead a difference of **lens, jurisdiction, state compression, aggregation level**, or only then **dynamics**. Different coordinates can describe the same predictive law. Different horizons can make different state variables sufficient. A genuine dynamics conflict begins only when the same state, lens and intervention imply different future records.

This is a programme rather than a theorem that every historical dispute will dissolve. Its proposed standard is nevertheless sharp: **a scientific disagreement is not fully posed until both sides have been brought onto a common empirical grammar.**

Consciousness on the map, not in the axioms

The formal observer used above need not be conscious. It is defined operationally by interventions and records; making consciousness an axiom would beg precisely the question a science of consciousness must investigate.

Human science, however, is practiced by conscious organisms. First-person experience is therefore not outside nature, and it should not disappear merely because third-person instruments are easier to formalize. In a science of consciousness, first-person reports or structured experiential records and third-person neural or behavioral records can be treated as distinct observational channels within a joint predictive problem. The question becomes: **what state closes their joint future law under intervention?**

Consciousness is returned to the map without being made a mystical cause. The framework neither claims that consciousness creates physical law nor treats the observer as an external contamination to be removed. It places every mode of access - including first-person access when scientifically operationalized - inside the same demand for predictive sufficiency, explicit jurisdiction and held-out discrimination.

Conclusion: The Observer and the World

We do not stand outside nature when we describe it.

We are parts of it, interrogating it with finite bodies, instruments, interventions and records. The first task of empirical science is therefore not to announce the world's nouns from a view from nowhere. It is to determine which distinctions available at an interface remain consequential to the future.

That gives the paper its first-principles order:

physical access $\rightarrow$ predictive closure $\rightarrow$ state $\rightarrow$ composition
 $\rightarrow$ observation geometry $\rightarrow$ reconstruction by failure.

A **state** is a compression of history. To call two histories the same state is to wager that no declared future experiment will ever need the distinction that was erased. A **law** is what remains well defined after that compression. A **geometry** records what the observational lens preserves. And when prediction fails, nature has returned a distinction the representation tried to forget.

Within that larger grammar, Theorem 1 gives one exact and unusually transparent observer geometry. If common scale is unavailable and the relevant homogeneous channels transform linearly, the empirical state is projective:

linear homogeneous state $\rightarrow$ Projective Shadow.

On the projective line, the finite law is fractional-linear, the continuous law is Riccati, and the cross-ratio is the invariant fingerprint. This structure is not implied by boundedness, saturation, or observation in general. It is earned by a specific lens. That is why the same Möbius grammar can recur in relativity, optics, networks, inference and selection without implying that those sciences share a substance: they can share an observational structure while describing different things.

The Empirical Grammar exists because nature need not remain in that branch. Predictive state can be stochastic. A scalar can fail to close. A point map can fail. Cross-ratios can fail. A hidden coordinate can matter only at a longer horizon. The correct response is not to protect the curve. It is to type the failure, enlarge the state when warranted, and demand that the new description predict futures it was not built to fit.

The synthetic study shows that this logic can work under controlled conditions and can fail in the intended direction under process noise, misspecification, correlated error, uneven sampling and weak state separation. The HepG2 experiment places the same architecture at genuine empirical risk. Its strongest possible outcome is not a beautiful fit. It is the sequence

earned fast closure $\rightarrow$ controlled failure $\rightarrow$ identified missing capacity $\rightarrow$ held-out recovery.

Its cleanest negative outcomes are equally valuable: rejection of the Point-Map Gate, rejection of the Projective Fingerprint, failure of the False-Noun test, or failure of the proposed enlarged state to recover prediction. The framework earns credibility only if those outcomes are allowed to kill the corresponding branch.

The same discipline changes how scientific disagreement should be approached. Two theories cannot be said to conflict cleanly while they use different observational lenses, different jurisdictions, different state compressions or different horizons. Once those are aligned, either the future laws coincide - in which case the disagreement is representational within that jurisdiction - or they separate, in which case the dispute has become an experiment.

And the observer itself is no longer absent from the picture. The mathematics does not require consciousness, but human knowledge is consciously lived and experimentally situated. A mature science should neither elevate that fact into metaphysical magic nor erase it in the name of objectivity. It should declare the channel, declare the jurisdiction, and ask what predictive distinctions survive.

Nature does not owe science its preferred nouns. A noun earns the name **state** only when the future can no longer distinguish the histories it compresses. A geometry earns physical meaning only when the lens that induces it has been identified and tested. A hidden dimension earns reality in the model only when restoring it recovers held-out law.

We are inside the world we measure. Begin there. Earn state. Test the lens. Let failure return what was forgotten. Let the future decide.

This is the intended meaning of a **grammar of empirical science**. A grammar does not dictate every sentence nature can speak. It specifies how valid descriptions are constructed, how apparently different descriptions can be recognized as equivalent, and how contradiction forces revision. The ambition of the programme is therefore not to impose one geometry on every science. It is to make explicit the conditions under which state, geometry and disagreement become scientifically meaningful at all.

If that standard survives contact with natural systems and unrelated scientific controversies, its reach will not come from claiming universality in advance. It will come from becoming difficult to do careful science without first asking the question on which this paper begins:

What can this observer distinguish, and which of those distinctions does the future still remember?

Appendix A: The Empirical Grammar in Practice

The Empirical Grammar places Theorem 1 inside the broader first-principles order of predictive closure, observer geometry and failure-driven reconstruction. It is placed in the appendix to keep the main text theorem-centred. The order matters because each later structure is licensed only after the earlier one has survived.

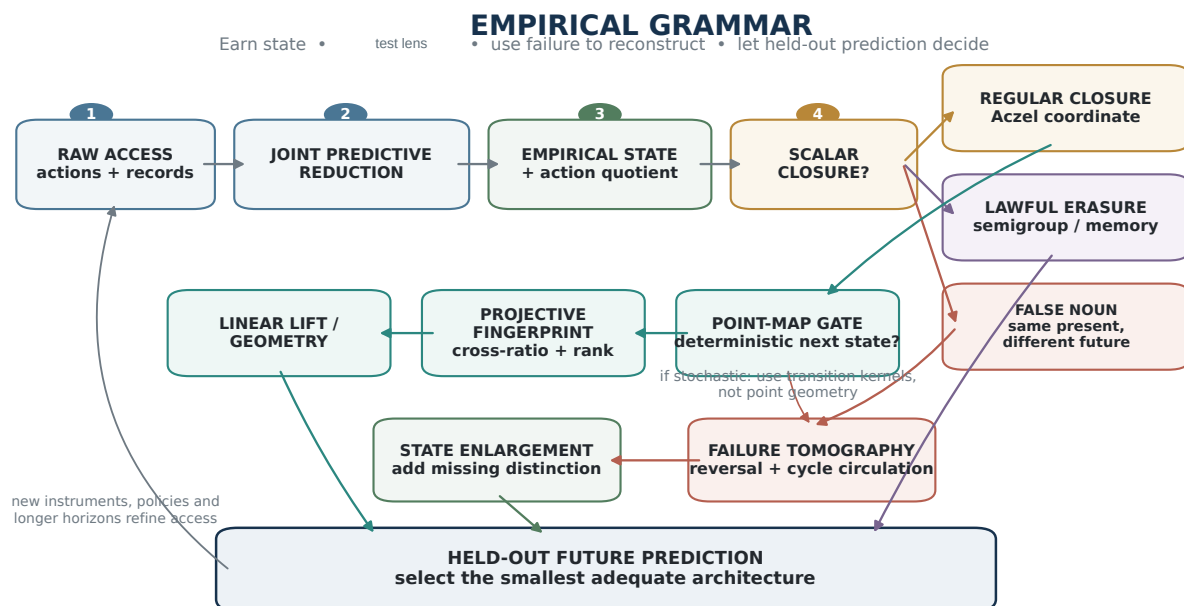


Figure 4: The Empirical Grammar. Physical access is reduced to predictive state and effective jurisdiction before coordinates or geometry are assigned. Scalar closure, the Point-Map Gate, the Projective Fingerprint, and projective rank are empirical tests. Failure feeds back through state enlargement, and every branch returns to held-out future prediction.

Step 1: Declare the physical access layer

List what the apparatus can actually do and record, together with the background assumptions needed to treat those actions and records as reproducible. The repertoire is theory-laden; the latent

state geometry is not assumed merely because the apparatus has coordinates.

Step 2: Learn predictive state and effective jurisdiction jointly

Estimate future-law features, reconstruct the coarsest supported history classes, and quotient interventions that are future-law indistinguishable. Include adaptive policies as probes when useful; the policy-closure proposition guarantees that the same predictive state notion applies.

Step 3: State the horizon, resolution and estimator

Report the future-test family, predictive metric, uncertainty method, clustering/model-selection rule and held-out split. Approximate closeness is not an equivalence relation.

Step 4: Test proposed state variables

If matched proposed states produce different future laws, the variable is not sufficient. Enlarge the state before assigning geometry.

Step 5: Test scalar sequential closure

- no single-valued composition: Case I, state failure;
- associative but nonregular/non-cancellative composition: Case II, lawful erasure/semigroup;
- continuous strictly monotone closure: Case III, reconstruct the Aczél natural coordinate $r = \phi(x)$.

Step 6: Pass the Point-Map Gate

Estimate the transition kernel $K_u(s, \cdot)$. If next predictive state retains non-degenerate stochastic spread beyond reconstruction error, reject deterministic point geometry for that representation. Model the kernel or enlarge state instead.

Step 7: Test the Projective Fingerprint Rather Than Fitting It

For a deterministic one-dimensional branch, test cross-ratio preservation on unused quadruples. A pass licenses a Möbius representation and an abstract two-channel homogeneous lift. A failure leaves a non-projective regular branch unless state adequacy itself has failed.

Step 8: Select projective dimension under penalty

When richer predictive coordinates exist, compare $\mathbb{P}^{d-1}$ candidates only up to a predeclared $d_{\max}$. Choose the smallest dimension supported by held-out prediction or a declared complexity penalty. Do not use unbounded rank as an unfalsifiable repair.

Step 9: Search for physical homogeneous channels

An abstract lift does not identify its physical coordinates. Ask whether independently measurable domain variables transform linearly/homogeneously in the required way. Failure to find them is a physical-lift failure, not a mathematical contradiction.

Step 10: Earn smooth calculus

Use Lie brackets only after the reconstructed predictive-law family has an identifiable smooth finite-dimensional constant-rank realization. Otherwise use discrete graph cycles and nonparametric predictive tests.

Step 11: Use Failure Tomography

Where smoothness holds, test the ab reversal law and iterated commutators. At finite amplitude or without smoothness, test graph-cycle circulation. Candidate new coordinates are hypotheses generated by the defect, not automatically real dimensions.

Step 12: Demand held-out recovery

Freeze the enlarged architecture and predict future records under histories/policies that were not used to construct it. A missing variable earns state status only if it improves held-out predictive law after complexity is accounted for.

What can actually fail?

The constitutive and empirical layers have different failure meanings. The existence of a quotient relative to stated conditional laws is mathematical. Its usefulness and every downstream scientific branch are exposed.

1. **Finite reconstruction failure:** learned states or action classes do not generalize.
2. **Policy failure:** equivalence appears open-loop but adaptive data violate the non-anticipating model assumptions or estimated conditional laws.
3. **Scalar-state failure:** matched scalar values produce different future laws.
4. **Aczél-regularity failure:** closure exists but continuity, strict monotonicity or cancellation fails.
5. **Deterministic-map failure:** $K_u(s, \cdot)$ remains non-degenerate, invalidating point geometry.
6. **Projective failure:** cross-ratios are not preserved beyond uncertainty.
7. **Finite-rank failure:** no candidate through $d_{\max}$ survives penalized held-out prediction.
8. **Physical-lift failure:** an abstract projective representation exists but proposed physical channels do not obey the required homogeneous linear dynamics.
9. **Smoothness failure:** predictive state is discrete, singular or non-constant-rank, invalidating Lie-bracket inference.
10. **Bracket/integrability failure:** reversal scaling or cycle structure rejects the proposed local/global reconstruction.
11. **State-recovery failure:** added variables do not improve held-out future laws.

The framework is therefore a **model-selection and state-reconstruction calculus**, not a universal physical law insulated from data. Its most general layer is predictive closure; the Projective Shadow is a conditional geometric branch.

Data and code availability

The complete synthetic demonstration is supplied as reproducible supplementary material: the simulation scripts, fixed random seeds, frozen intermediate tables, held-out predictions, robustness outputs, reversal-scaling data, weak-separation stress test, and figure-generation inputs are provided together so that the reported numerical results can be regenerated and audited. The same package is intended as the reference implementation for applying the workflow to new domains. The HepG2 protocol is prospective; confirmatory analysis code, exclusion rules, equivalence margins and held-out splits are to be frozen before outcomes are opened.

References

1. Shalizi, C. R., & Crutchfield, J. P. (2001). Computational mechanics: Pattern and prediction, structure and simplicity. *Journal of Statistical Physics*, 104, 817-879. Preprint: arXiv:cond-mat/9907176.
2. Barnett, N., & Crutchfield, J. P. (2015). Computational mechanics of input-output processes: Structured transformations and the epsilon-transducer. *Journal of Statistical Physics*, 161, 404-451. doi: 10.1007/s10955-015-1327-5.
3. Littman, M. L., Sutton, R. S., & Singh, S. (2002). Predictive representations of state. In *Advances in Neural Information Processing Systems 14*.
4. Aczél, J. (1966). *Lectures on Functional Equations and Their Applications*. Academic Press.
5. Aczél, J. (1989). *A Short Course on Functional Equations*. D. Reidel/Springer.
6. Schiff, J., & Shnider, S. (1999). A natural approach to the numerical integration of Riccati differential equations. *SIAM Journal on Numerical Analysis*, 36, 1392-1413.
7. Bor, G., Levi, M., Perline, R., & Tabachnikov, S. (2020). Tire tracks and integrable curve evolution. *International Mathematics Research Notices*, 2020(9), 2698-2768.
8. Ungar, A. A. (1988). The Thomas rotation formalism underlying a nonassociative group structure for relativistic velocities. *Applied Mathematics Letters*, 1, 403-405.
9. Ungar, A. A. (1989). The relativistic noncommutative nonassociative group of velocities and the Thomas rotation. *Results in Mathematics*, 16, 168-180.
10. Barriuso, A. G., Monzon, J. J., Sanchez-Soto, L. L., & Carinena, J. F. (1999). Lossless multilayers and Lorentz transformations: more than an analogy. *Optics Communications*, 162, 1-6. doi: 10.1016/S0030-4018(99)00065-6.
11. Monzon, J. J., & Sanchez-Soto, L. L. (1999). Origin of the Thomas rotation that arises in lossless multilayers. *Journal of the Optical Society of America A*, 16, 2786-2792. doi: 10.1364/JOSAA.16.002786.

12. Sanchez-Soto, L. L., Monzon, J. J., Barriuso, A. G., & Carinena, J. F. (2012). The transfer matrix: a geometrical perspective. *Physics Reports*, 513, 191-227.
13. Pasterski, S., Shao, S.-H., & Strominger, A. (2017). Flat space amplitudes and conformal symmetry of the celestial sphere. *Physical Review D*, 96, 065026. arXiv:1701.00049.
14. Kogelnik, H., & Li, T. (1966). Laser beams and resonators. *Applied Optics*, 5, 1550-1567. doi: 10.1364/AO.5.001550.
15. Gupta, M. S. (2006). Escher's art, Smith chart, and hyperbolic geometry. *IEEE Microwave Magazine*, 7(5), 66-76. doi: 10.1109/MW-M.2006.247916.
16. Muller, A. A., et al. (2018). A review and mathematical treatment of infinity on the Smith chart, 3D Smith chart and hyperbolic Smith chart. *Symmetry*, 10, 458. doi: 10.3390/sym10100458.
17. Orr, H. A. (2009). Fitness and its role in evolutionary genetics. *Nature Reviews Genetics*, 10, 531-539. doi: 10.1038/nrg2603.
18. Svensson, E. I., & Connallon, T. (2019). How frequency-dependent selection affects population fitness, maladaptation and evolutionary rescue. *Evolutionary Applications*, 12, 1243-1258.
19. Murray, D. J. (2026). Hormesis as a geometric necessity of bounded adaptive systems. *Dose-Response*. doi: 10.1177/15593258261469171.
20. Murray, D. J. (2026). A dynamical model of glutathione homeostasis in G6PD deficiency and NRF2-activated non-small cell lung cancer. *Redox Biochemistry and Chemistry*, 17, 100084. doi: 10.1016/j.rbc.2026.100084.
21. Ferns, N., Panangaden, P., & Precup, D. (2004). Metrics for finite Markov decision processes. In *Proceedings of the 20th Conference on Uncertainty in Artificial Intelligence*, 162-169.
22. Abate, A. (2013). Approximation metrics based on probabilistic bisimulations for general state-space Markov processes: a survey. *Electronic Notes in Theoretical Computer Science*, 297, 3-25. doi: 10.1016/j.entcs.2013.12.002.
23. D'Alessandro, D. (1998). Invariant manifolds and projective combinations of solutions of the Riccati differential equation. *Linear Algebra and its Applications*, 279, 181-193. doi: 10.1016/S0024-3795(98)00013-5.
24. Shalizi, C. R., & Shalizi, K. L. (2004). Blind construction of optimal nonlinear recursive predictors for discrete sequences. In *Proceedings of the 20th Conference on Uncertainty in Artificial Intelligence*, 504-511. arXiv:cs/0406011.
25. Boots, B., Siddiqi, S. M., & Gordon, G. J. (2011). Closing the learning-planning loop with predictive state representations. *The International Journal of Robotics Research*, 30(7), 954-966. doi: 10.1177/0278364911404092.
26. Siddiqi, S., Boots, B., & Gordon, G. J. (2010). Reduced-rank hidden Markov models. *Proceedings of Machine Learning Research*, 9, 741-748.
27. Gutscher, M., Pauleau, A.-L., Marty, L., Brach, T., Wabnitz, G. H., Samstag, Y., Meyer, A. J., & Dick, T. P. (2008). Real-time imaging of the intracellular glutathione redox potential. *Nature Methods*, 5, 553-559. doi: 10.1038/nmeth.1212.
28. Morgan, B., Sobotta, M. C., & Dick, T. P. (2011). Measuring E(GSH) and H₂O₂ with roGFP2-based redox probes. *Free Radical Biology and Medicine*, 51, 1943-1951. doi: <https://doi.org/10.1016/j.free-radbiomed.2011.08.035>.
29. Deferme, L., Briedé, J. J., Claessen, S. M. H., Jennen, D. G. J., Cavill, R., & Kleinjans, J. C. S. (2013). Time series analysis of oxidative stress response patterns in HepG2: A toxicogenomics approach. *Toxicology*, 306, 24-34. doi: <https://doi.org/10.1016/j.tox.2013.02.001>. GEO: GSE39291.
30. Ghergurovich, J. M., García-Cañaveras, J. C., Wang, J., et al. (2020). A small molecule G6PD inhibitor reveals immune dependence on pentose phosphate pathway. *Nature Chemical Biology*, 16, 731-739. doi: <https://doi.org/10.1038/s41589-020-0533-x>.
31. Tu, M., Fan, X., Shi, J., Jing, S., Xu, X., & Wang, Y. (2022). 2-Fluorofucose attenuates hydrogen peroxide-induced oxidative stress in HepG2 cells via Nrf2/keap1 and NF-κB signaling pathways. *Life*, 12, 406. doi: 10.3390/life12030406.
32. Erler, K., Krafczyk, N., Steinbrenner, H., & Klotz, L.-O. (2024). Selective activation of cellular stress response pathways by fumaric acid esters. *FEBS Open Bio*, 14, 1230-1246. doi: 10.1002/2211-5463.13833.